

Introduction to DEGAS

Single-cell RNA sequencing (scRNA-seq) has transformed our ability to resolve cellular heterogeneity within complex tissues, revealing distinct cell types and transcriptional states that drive biological processes and disease. In contrast, most large clinical cohorts still rely on bulk transcriptomic profiling, where gene expression reflects an averaged signal across mixed cell populations. This mismatch makes it difficult to connect patient-level disease phenotypes to specific cellular drivers.

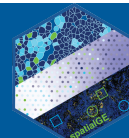
Diagnostic Evidence GAuge of Single cells (DEGAS) addresses this challenge through a deep transfer learning framework that integrates scRNA-seq data with bulk patient transcriptomes. By jointly modeling these complementary data types, DEGAS enables the transfer of disease-associated signals from patient samples to individual cells.

DEGAS uses a multitask neural network that learns a shared latent representation of gene expression across bulk and single-cell datasets. During training, the model simultaneously optimizes patient-level prediction tasks—such as disease subtype, risk group, or survival outcome—while incorporating single-cell profiles that capture cellular diversity. Shared hidden layers allow the network to extract common biological structure and propagate disease-relevant patterns from bulk data into the single-cell space.

This approach yields disease association scores for individual cells, quantifying how closely each cell's expression profile aligns with disease-related transcriptional signatures observed in patient cohorts. As a result, DEGAS can highlight cell types or states that may contribute to disease progression, even when clinical datasets lack single-cell resolution. The framework is flexible and supports a range of clinical labels and outcomes. By bridging patient-level transcriptomics with single-cell biology, DEGAS provides a powerful strategy for identifying disease-relevant cellular populations and uncovering mechanisms underlying complex diseases.

Reference:

Johnson TS, Yu CY, Huang Z, Xu S, Wang T, Dong C, Shao W, Zaid MA, Huang X, Wang Y, Bartlett C, Zhang Y, Walker BA, Liu Y, Huang K, Zhang J. Diagnostic Evidence GAuge of Single cells (DEGAS): a flexible deep transfer learning framework for prioritizing cells in relation to disease. *Genome Med.* 2022;14(1):11. Epub 2022/02/03. doi: 10.1186/s13073-022-01012-2. PubMed PMID: 35105355; PMCID: PMC8808996.



DEGAS

Method description...

Bulk-RNA expression study ?

Clinical feature ?

1 BRCA

2 SUBTYPE

Risk categories ?

Non-risk categories ?

3 BRCA_Basal (171) x

4 BRCA_Normal (36) x

5 Threshold for removing non-expressed genes in the study: 0.3 ?

6 Top variable genes percentile: 0.2 ?

7 Cutoff for adjusted P-value in TCGA DE analysis: 0.05 ?

8 Number of layers in the neural network ? 3

9 Number of Bootstrap samples ? 5

10 Annotation to test ? STclust; Domains (k): 02; spatial weight:0.02

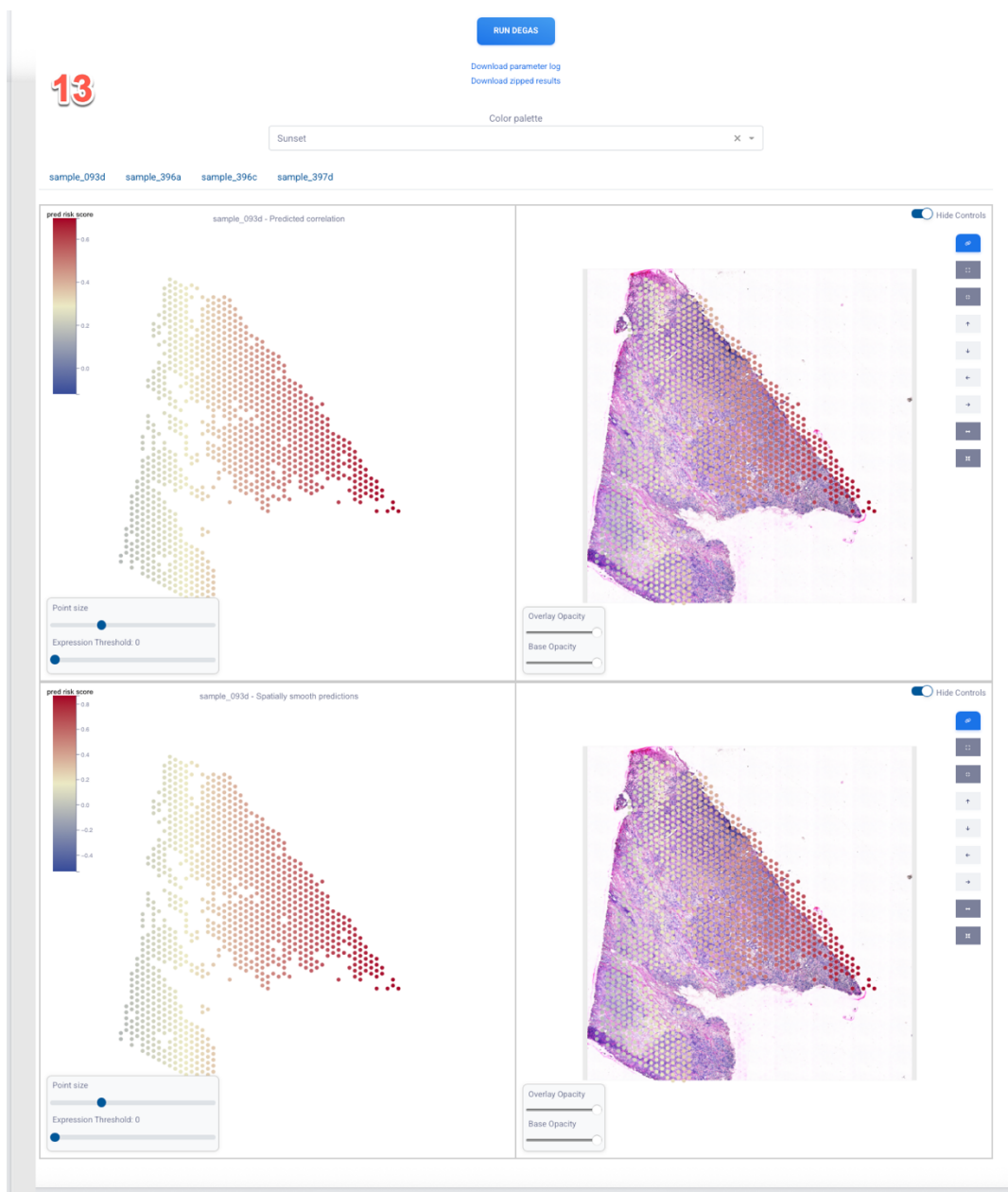
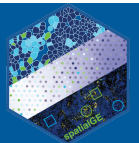
RUN DEGAS 12

[Download parameter log](#)

[Download zipped results](#)

11 Color palette Sunset

sample_093d sample_396a sample_396c sample_397d

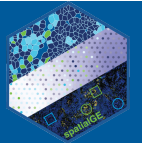


DEGAS Analysis Documentation:

Integration of Visium Breast Cancer Spatial Data with TCGA-BRCA Bulk RNA-seq

The **DEGAS (Diagnostic Evidence GAUGE of Single cells)** framework was used to integrate **spatial transcriptomics data from breast cancer Visium samples** with **bulk RNA-seq data from the TCGA-BRCA cohort**. DEGAS applies deep transfer learning to propagate disease signals learned from patient bulk transcriptomes to individual spatial spots (treated as pseudo-cells).

The following describes the configuration and execution steps used in the DEGAS interface.



Step-by-Step DEGAS Configuration

Step 1 – Bulk RNA Expression Dataset

Dataset Selected:

BRCA (TCGA Breast Invasive Carcinoma)

This dataset provides **bulk RNA-seq expression profiles from breast cancer patients** in The Cancer Genome Atlas (TCGA). These data are used to learn **disease-associated transcriptional patterns**.

#For this demo following clinical features are chosen. Choose clinical feature based on your study.

Step 2 – Clinical Feature

Clinical variable:

SUBTYPE

The **molecular subtype classification** of breast cancer from TCGA was used as the clinical label for training the model.

Examples of subtypes include:

- Luminal A
- Luminal B
- HER2-enriched
- Basal-like

The neural network learns gene expression signatures associated with these subtypes.

Step 3 – Risk Categories

High-risk category selected:

BRCA Basal

Basal-like tumors are typically associated with **more aggressive disease behavior and poorer prognosis**, and were therefore defined as the **risk group** for the DEGAS model.

Step 4 – Non-Risk Categories

Non-risk category selected:

BRCA Normal

Samples labeled as normal or non-aggressive serve as the **reference class** during model training.

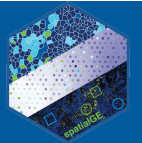
The DEGAS model therefore learns to distinguish **Basal tumor signatures from non-risk expression profiles**.

Step 5 – Threshold for Removing Unexpressed Genes

Parameter: 0.3

Genes with expression values below this threshold across samples are removed before training.

Purpose:



- Reduce noise
- Improve model stability
- Remove genes with very low expression

Step 6 – Top Variable Genes Percentile

Parameter: 0.2 (top 20%)

Only the **top 20% most variable genes** are retained for model training.

Highly variable genes capture:

- biologically relevant transcriptional differences
- disease-specific signals

Step 7 – TCGA Adjusted p-value Cutoff

Parameter: 0.05

Genes significantly associated with the clinical feature (SUBTYPE) in TCGA are filtered using an **adjusted p-value threshold of 0.05**.

This step enriches the model for **genes most relevant to breast cancer subtype classification**.

Step 8 – Neural Network Depth

Parameter: 3 layers

The DEGAS deep learning model contains **three hidden layers**.

This architecture enables the model to:

- learn nonlinear gene expression relationships
- capture shared latent features between bulk and spatial data.

Step 9 – Bootstrap Samples

Parameter: 5

Bootstrapping is used to estimate the **stability of DEGAS predictions**.

Multiple models are trained on resampled datasets, and the final score reflects **consensus predictions across bootstrap iterations**.

Step 10 – Annotation to Test

Selection:

STclust Domains (K=12 spatial weight=0.2)

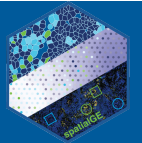
Spatial domain annotations generated from clustering analysis are used for **testing DEGAS predictions**.

This allows the evaluation of whether **specific spatial domains are enriched for disease-associated signals**.

Step 11 – Spatial Transcriptomics Input Data

Input samples: Visium breast cancer samples

Example samples:



- sample_394A
- sample_394B
- sample_394C
- sample_397A

Each Visium spot represents a **pseudo-cell** containing spatially resolved transcriptomic data.

Step 12 – Run DEGAS

After all parameters are configured, the **RUN DEGAS** button executes the model.

The DEGAS pipeline performs:

1. Gene filtering
2. Differential expression analysis in TCGA
3. Neural network training
4. Transfer learning between bulk and spatial data
5. Calculation of **disease relevance scores for each spatial spot**

Step 13 – Output Visualization

The output consists of **spatial DEGAS score maps** for each Visium sample.

Two major visualizations are produced:

1. DEGAS Score Map

Each spatial spot is assigned a **DEGAS disease score**, representing how strongly its transcriptional profile resembles the **Basal breast cancer subtype signature learned from TCGA**.

Color scale:

- **Red** → strong disease association
- **White/neutral** → intermediate
- **Blue** → weak disease association

2. Spatial Overlay Map

DEGAS scores are overlaid on the **spatial tissue architecture**, allowing visualization of where disease-associated transcriptional patterns occur within the tissue section.

This enables identification of:

- tumor regions
- microenvironment niches
- spatial heterogeneity of disease signals

Note:

DEGAS requires ≥ 2 domains. Samples will be skipped if they have 1 domain and less than 50 cells.